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Java SE 27 & JDK 27

Interface Enumeration<E>

Type Parameters:

E - the type of elements returned by this enumeration

All Known Subinterfaces:

NamingEnumeration<T>

All Known Implementing Classes:

StringTokenizer

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public interface **Enumeration<E>**

An object that implements the Enumeration interface generates a series of elements, one at a time. Successive calls to the nextElement method return successive elements of the series.

For example, to print all elements of a Vector<E> *v*:

```
for (Enumeration<E> e = v.elements(); e.hasMoreElements();)
    System.out.println(e.nextElement());
```

Methods are provided to enumerate through the elements of a vector, the keys of a hashtable, and the values in a hashtable. Enumerations are also used to specify the input streams to a SequenceInputStream.

API Note:

The functionality of this interface is duplicated by the Iterator interface. In addition, Iterator adds an optional remove operation, and has shorter method names. New implementations should consider using Iterator in preference to Enumeration. It is possible to adapt an Enumeration to an Iterator by using the asIterator() method.

Since:

1.0

See Also:

Iterator, SequenceInputStream, nextElement(), Hashtable, Hashtable.elements(), Hashtable.keys(), Vector, Vector.elements()

Method Summary

All Methods	Instance Methods	

Modifier and Type	Method	Description
default Iterator<E>	**asIterator**()	

		Returns an `Iterator` that traverses the remaining elements covered by this enumeration.
`boolean`	**`hasMoreElements`**`()`	Tests if this enumeration contains more elements.
`E`	**`nextElement`**`()`	Returns the next element of this enumeration if this enumeration object has at least one more element to provide.

Method Details

hasMoreElements

`boolean hasMoreElements()`

Tests if this enumeration contains more elements.

Returns:

`true` if and only if this enumeration object contains at least one more element to provide; `false` otherwise.

nextElement

`E nextElement()`

Returns the next element of this enumeration if this enumeration object has at least one more element to provide.

Returns:

the next element of this enumeration.

Throws:

`NoSuchElementException` - if no more elements exist.

asIterator

`default Iterator<E> asIterator()`

Returns an `Iterator` that traverses the remaining elements covered by this enumeration. Traversal is undefined if any methods are called on this enumeration after the call to `asIterator`.

API Note:

This method is intended to help adapt code that produces `Enumeration` instances to code that consumes `Iterator` instances. For example, the `JarFile.entries()` method returns an `Enumeration<JarEntry>`. This can be turned into an `Iterator`, and then the `forEachRemaining()` method can be used:

```
JarFile jarFile = ... ;
jarFile.entries().asIterator().forEachRemaining(entry -> { ... });
```

(Note that there is also a `JarFile.stream()` method that returns a `Stream` of entries, which may be more convenient in some cases.)

Implementation Requirements:

The default implementation returns an `Iterator` whose `hasNext` method calls this Enumeration's `hasMoreElements` method, whose `next` method calls this Enumeration's `nextElement` method, and whose `remove` method throws `UnsupportedOperationException`.

Returns:

an Iterator representing the remaining elements of this Enumeration

Since:

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Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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